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Wings on the Ground:
How can we provide our abandoned airplanes a second chance at life?



Figure 1: Old, abandoned airplane vs. Soaring Airplane: What's the difference?

Recently, I went on a flight from Austin to Newark. It was my first time sitting away from my family, and my first time sitting with two other strangers. Naturally, I was nervous. But, one of the passengers in my row began introducing herself to both of us, telling us what she does and where she is from. The passenger next to me and I introduced ourselves. Interesting. Weird, but interesting. I think it was only sheer destiny and fate that told me to sit next to a Postdoctoral fellow in English Literature and an aerospace engineer.

I have to admit, the Postdoctoral fellow was an environmentally conscious person. She made sure she had the most sustainable versions of the items she needed possible. So, she asked the aerospace engineer, "How do companies in the aerospace industry make sure that the work

they're doing and the parts that are being made are environmentally sustainable? Do they use recycled materials at all?" The aerospace engineer replied, "Currently, there isn't much use of recycled materials in airplane parts. You see, most of the materials that make up airplane parts, like the covering of the airplane engines there, are composite materials. But, companies are very hesitant to adopt methods to recycle composite materials because the methods are very expensive and complex."

This statement struck me. What methods are so hard and expensive that not even the big aerospace companies, like Boeing or Airbus, not adopting? Why are they so hard and expensive? First of all, why should we even care about recycling composite materials?

Now, imagine another story. A story of the abandoned airplanes. Imagine an abandoned airplane in an uninhabited plot of land. Imagine this same abandoned airplane with the other abandoned airplanes, all on the same plot. Now, imagine all the airplanes left abandoned in lots similar to the one we just imagined. Imagine all of the lots of airplanes across the world. Chipping and rusting away and their paint and lettering fading, the airplanes are covered in piles and piles of dust. No one is present to care for and fly them.

When we look back at the history of such airplanes, we can see that they've experienced their own ups and downs. They've had their glory days; they've undoubtedly had their prime. But, a few issues arise, and these airplanes are destined to be sent to the graveyard.

Yet, is this fair? For what reasons are these airplanes being left behind indefinitely? On what basis do companies decide whether airplanes will be abandoned or not?

A few reasons why airplanes might be abandoned are because of higher fuel costs, negative revenue streams for the companies that own the airplanes, and world events ("Factors Involved in Fleet Retirement and Aircraft Recycling"). According to Johnnie Norsworthy, an aerospace administration and operations major at Oklahoma State University, because of COVID-19, "[w]ith demand falling massively below what it used to be and supply remaining relatively the same...[a]irlines have already slashed their schedules to match the reduced demand for air travel, leaving as many as hundreds of aircraft per airline without any scheduled flights." In some cases, these aircraft can be flown occasionally so that they will be up-to-date with maintenance. If they are grounded, then the aircraft will have to go through "lengthy maintenance evaluations before they were allowed to fly again" (Norsworthy). Some could have malfunctioning parts, which negatively impacts the operations of the whole airplane. And some can be replaced by airline companies to hold more people on each flight; to offer better amenities to passengers, in the hopes that the increase in passenger experience will cause an increase in revenue; or to reduce operational costs ("Navigating shifting skies: The evolving landscape of the airline industry").

For whatever reason these airplanes are abandoned, maybe there is a use for them. If they will not be flown indefinitely, maybe we can incorporate them into our journey on the evolution of aircraft. Let me explain.

In the modern world, as countries are becoming more interconnected with each other, travel by air has become a very popular option, especially in the case of interstate or international travel. Airplanes are appealing to people because of the comfort, lower times, or both. But, because of this overall rise in demand for travel by air, the demand of new airplanes that meet the evolving standardized regulations and consumer expectations for aircraft is rising. As a result,

companies are producing higher quantities of airplanes that meet the evolving needs of their consumers, to potentially increase the revenue and profit that airline companies can receive.

With this rise in demand for new airplanes, companies have been needing to produce more parts. Creating each part from scratch will take considerable amounts of time, money, and resources. But, what if we can cut down on the amount of time, money, and resources for creating airplane parts?

Like all other materials made on Earth, airplane parts are composed of natural resources, especially metals. But there are only so many natural resources on Earth to fulfill all our needs. Since the parts in the abandoned airplanes are also made of similar, if not the same, parts and materials, the parts of the abandoned airplanes could, in theory, be reused or broken down into its individual components. These individual components can contribute to the production of new parts, should they be reused or recycled.

Now, even if some of the parts can be reused or recycled, there are some parts which cannot. According to Farah A. Genena and other authors of the source, all researchers from the Mechanical Engineering Department of the United Arab Emirates University, these parts are often non-biodegradable and, when left in landfills, have the potential to ruin the environment surrounding where the parts are left.

Before we dive into the intricacies of the matter, let's first understand the structure and potential recycling methods of "[m]ost of the aerospace waste[:] pre-preg (or pre-impregnated) carbon fiber - which is non-biodegradable. Pre-preg means that the fibers are combined with a pre-defined amount of resin and is covered with a protective thin layer of polyethylene (more commonly known as nylon)" (Genena et al). Because of the composition of pre-preg carbon fibers, pre-preg carbon fibers are classified as composite materials. Composite materials are, for the most part, non-separable; according to Dr. Yongxiang Yang, an associate professor at the Technische Universiteit Delft, and the other authors of the source, it is very hard to recycle them because of "their inherent heterogeneous nature of the matrix and reinforcement." Therefore, in order to recycle composite materials, according to Dr. Eylem Asmatulu, a member of the Department of Industrial and Manufacturing Engineering at Wichita State University, and the other authors of the source, companies are using "detangling, cutting and hammer milling processes to remanufacture carbon fiber scraps for different product applications." So far, there are only three categories of recycling composite materials, each with its own shortcomings and advantages:

Mechanical recycling involves shredding and grinding followed by screening to separate fibre-rich and resin-rich fractions for re-use. The method is very energy-intensive and the recyclates have relatively low quality. *Thermal processing* uses high temperature (between 300 and 1000 °C) to decompose the resin and separate the reinforcement fibres and fillers. Clean fibres or inorganic fillers are re-generated, and secondary fuel or thermal energy can be produced through pyrolysis, gasification or combustion. However, the quality of the recovered fibres or filler materials degrades to a varying extent during thermal processing. *Chemical recycling* aims at chemical depolymerisation or removal of the matrix and liberation of fibres for further recycling by using organic or inorganic solvent. Lack of flexibility and generation of waste chemicals with environmental

concerns in chemical recycling caused the situation in which there is no active development at the moment (Yang et al).

Because of how each category of recycling composite materials requires labor-intensive methods, expensive equipment, and produces materials that are not yet comparable to virgin materials, aerospace companies are very hesitant to adopt methods to recycle composite materials, especially pre-preg carbon fibers.

But, while recycling pre-preg carbon fibers and other composite materials used to make airplane parts has its challenges, it also has its benefits. One of the main benefits is that it cuts down on the energy necessary to produce each part. According to Dr. Asmatulu and the other authors of the source, “[t]he energy consumption of the recycling process is 5% of that required for first-generation aluminum production.” While this example might apply only to recycled aluminum sheets, using recycled materials in producing new parts has seen tremendous decreases in energy necessary to produce each part. Furthermore, by using recycled materials in parts production, companies would not need to spend as much time, money, and resources when compared to the amount used in making virgin parts. There are also socioeconomic benefits to recycling composite materials, like pre-preg carbon fibers (Genena et al). Because the market for recycled airplane parts is increasing and more companies are dedicated to researching ways to make the process of recycling airplane parts easier and more adaptable to supply chains, recycling airplane parts is becoming a lucrative industry. For example, “[t]he recycled material is likely to be utilized for noncritical structures of aircraft, such as galleys, interior linings, seat parts, and tools that produce stronger and lighter-weight materials in the same industry” and “that the reclaimed fibres serve as a viable replacement for new fibre in many high-end industrial manufacturing processes, and [offer] a significant savings of money and carbon dioxide” (Asmatulu et al; Yang et al). Hence, with the current potential recycled airplane parts have to be used for many cases and in the aerospace industry, more companies and individuals are interested in the recycled airplane parts industry. Therefore, there are more jobs available in the industry, thus boosting the socio-economic level of the people who choose to work in the industry.

So while recycling composite materials, especially pre-preg carbon fibers, may have numerous benefits, the fact remains that it is equally hard to adopt methods to recycle composite materials. Because the aerospace industry is currently booming and there is a consistently high demand for airplanes, airplane companies are finding it necessary to speed up the amount of time taken to produce the individual parts for the airplane and to assemble and build airplanes.

Now, let's consider what would happen if airplane companies started adopting methods to recycle composite materials to be implemented into airplane production. In regards to our lives, when we do anything for the first time, we are usually slower at it because of how we make slight errors or are not comfortable with the process of doing it. Similarly, in the case of the airplane companies, because adopting methods to recycle composite materials will be a process that most airplane companies are doing for the first time, the companies will slow down the rate at which they produce each part. Afterwards, because of the slower rate at which parts are being produced, airplane assembly will also be delayed. Hence, airlines will see higher waiting times for their aircraft.

Because adopting methods to and using recycled composite materials will be a first time for many companies in the aerospace industry, the resulting product might not entirely fit the needs of aerospace companies. Therefore, companies will need to perfect methods to recycle composite materials to be used in their manufacturing and production processes. Companies will need more time, even though the time they need might not always be available.

For the time being, if recycled composite materials are used in the production for new airplanes, if they are used especially in the structural components of the airplanes, the airplanes may have some structural faults as the recycled composite materials may not be able to overcome or match the same conditions as virgin materials can. While on flight, should something happen because of the use of recycled composite materials, the lives of many could be jeopardized. There may be certain risks that are involved by releasing new airplanes using recycled composite materials in their structure. But, given the amount of parts needed by the aerospace industry, and how we may not have enough resources in the future to continue producing aircraft parts, we need to start seeing recycled airplane parts as a viable option. While it might not be ready to be released to the public, we need to begin testing to see how it might cope with the real world. We may need to add modifications and start researching how to better integrate recycled airplane parts into most or all aspects of the airplane, as a result.

But all this takes money and time, especially the money and time that companies, for the most part, are not willing to put. They might see it as something with too much risk or something that would cause their profits to decrease. Some companies, like Boeing and Airbus, have started working towards making airplanes more sustainable by using recycled airplane parts and working with companies that recycle composite materials. Yet, it is not a widespread practice. In the United States, there are very few regulations regarding what materials are being used while constructing airplanes. Most of the regulations that regard airplane operations and manufacturing look at whether the airplane is safe to fly and functional, and the parts and manufacturing processes conform to standards. As long as the airplane-manufacturing companies follow the Federal Aviation Administration and Department of Transportation's regulations to the bare minimum possible, their planes will be able to fly in the skies.

Companies, in most cases, do not have the incentive to start using recycled or reused airplane parts. In the case of reused airplane parts, companies will have to dismantle airplanes and individually categorize reusable and non-reusable parts (Asmatulu et al). In the case of recycled airplane parts, companies will have to adopt time-consuming, labor-heavy, expensive methods that yield materials that are not comparable to that of virgin materials. Companies might think it is easier to continue using the methods they already have.

We might think that companies might've an incentive to adopt methods for recycling and reusing airplane parts once we convince them about how the benefits outweigh the losses. But, we wouldn't know the extent to which they actually adopt methods, if they have at all. Furthermore, according to Elie Mystal, the justice correspondent at the news website *The Nation*, after the passage of the Airline Deregulation Act of 1978, airline companies were deregulated so that they could focus on efficiency and lowering the costs, as well as lowering fares for consumers. If airline companies are deregulated, the extent to which aerospace companies are regulated may be sufficient for the bare-minimum of protections. In many cases, companies bribe or pay government officials or organizations so that they can continue to do what they would

like. In each case, “[l]abor gets hollowed out, monopolies emerge, service gets worse, and consumer protections disappear. But prices stay low and the stock market goes up, so everybody acts like we’re winning.” But, while constructing airplanes, we have to make sure that they are absolutely safe to fly in.

Hence, in order for companies to adopt methods to recycle composite materials, the government should start creating regulations on what materials airline companies can use to construct airplanes, how much percentage of the airplane materials are from recycled or reused parts, and testing airplanes made out of majority recycled or reused parts. While some companies have already started using recycled airplane parts, others haven’t. For those who have, we have to incentivize them to continue; for those who haven’t, we need to encourage them to start.

Without the government incentive, it would be hard to attract companies’ interest toward recycled and reused airplane parts. It would be hard to convince companies to use the aircraft they abandoned. But, once the companies start, the aerospace industry will finally start their own path to becoming sustainable.

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